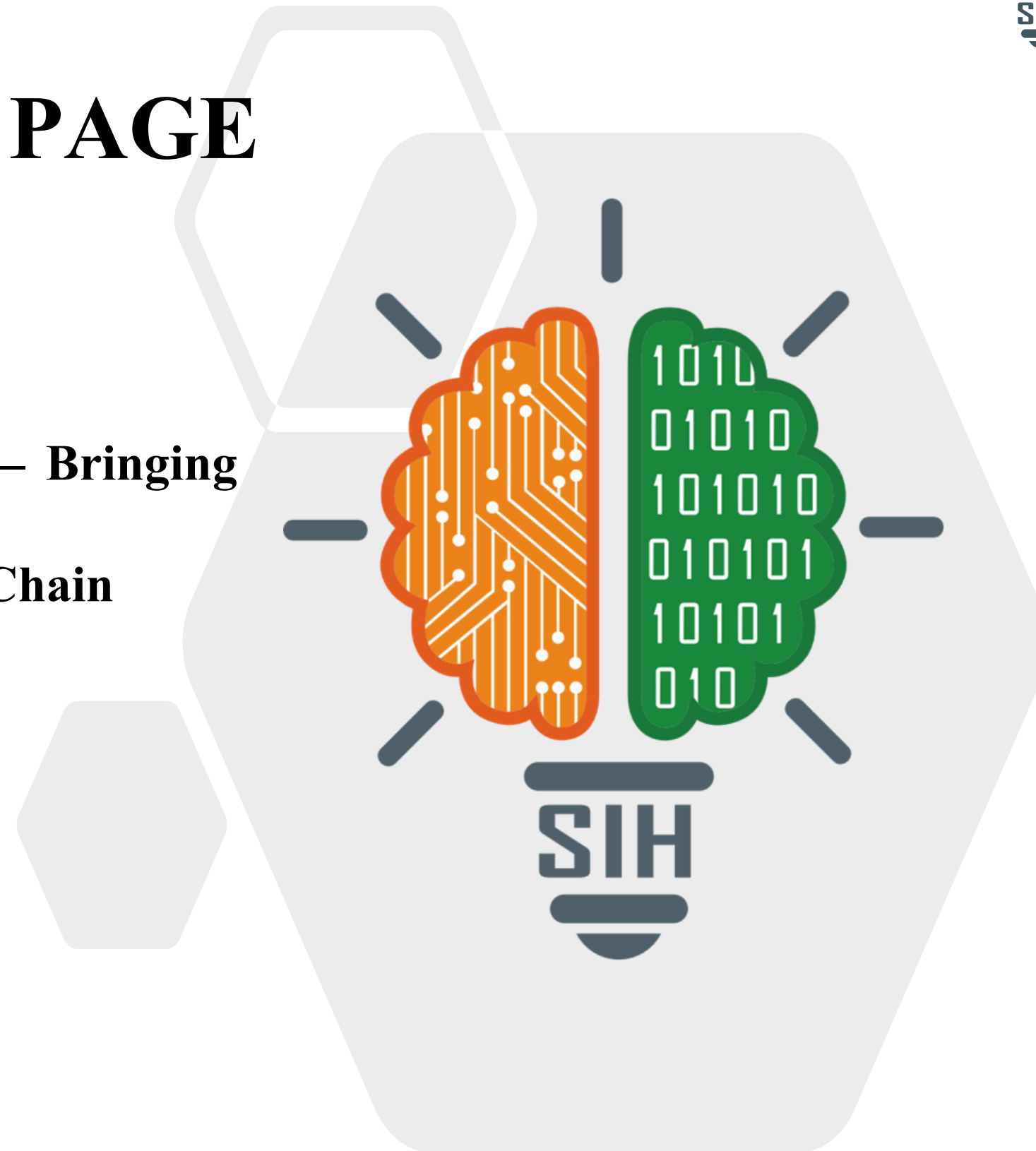


TITLE PAGE

- Problem Statement ID –26229
- Problem Statement Title - Kabadiwala Connect – Bringing the Informal Collector into the Formal Recycling Chain
- Theme - Clean & Green Technology
- PS Category- Software
- Team ID-
- Team Name (Registered on portal):Raddi Rangers



**Kabadiwala Connect — a two-tier digital bridge from informal collection to formal,
EPR-compliant recycling**

Kabadiwala Connect is a vernacular, offline-first mobile app and web portal that connects informal scrap pickers directly with authorized e-waste recyclers—using on-device AI for instant price discovery, automated neighborhood pickup batching, and tamper-proof digital handover receipts for government EPR compliance.

A voice-first price & safety companion for individual kabadiwalas — check fair rates and safe-handling guidance in Hindi/Marathi, even with zero literacy.

A full offline-first lot-and-handover app for scrap dealers/aggregators — the ones who actually have the volume to reach a recycler

Not replacing the informal chain — formalizing the transaction that's already happening, starting at the layer with the least visibility today

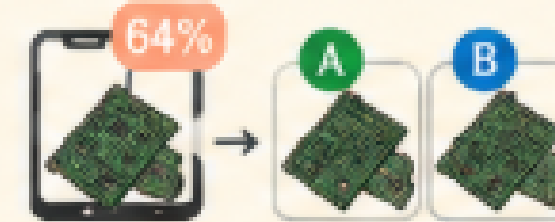


5 INNOVATIVE FEATURES

Real challenges.
Tangible change.

01

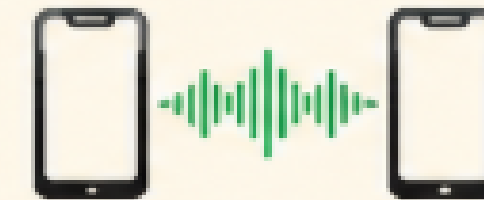
Confidence-Aware AI Fallback



If confidence is low (<75%), triggers a visual A/B question with voice prompts.

02

Acoustic Co-Location Handshake (The Chirp)



Proves collector and recycler are physically together, even in cell dead zones.

03

GPS-Pattern Anomaly Detection



Flags suspicious behavior if the same GPS location generates scrap lots under multiple collector IDs.

04

Vernacular Hazard & Safety Guidance



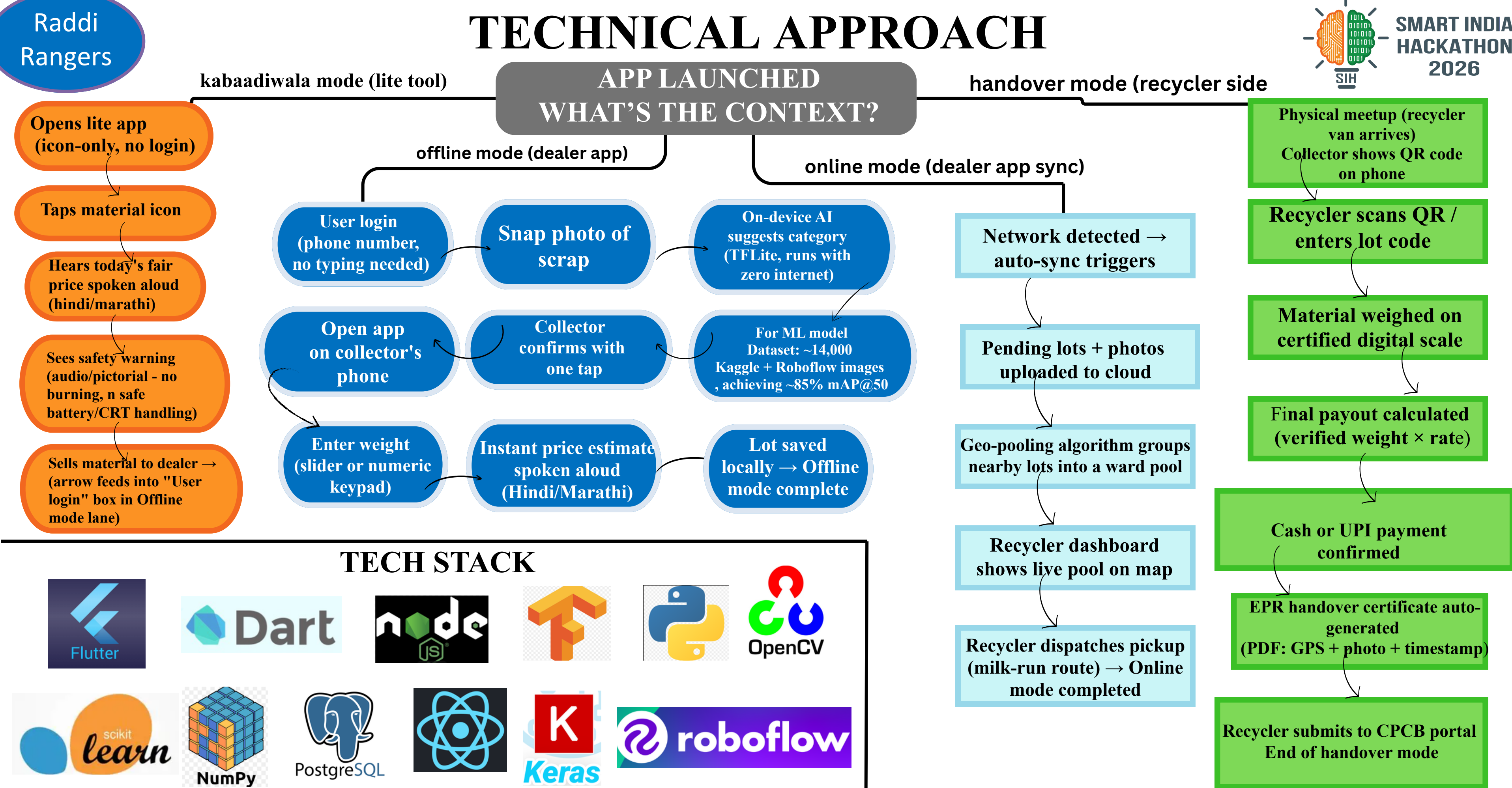
Audio and pictorial warnings built directly into the workflow to stop hazardous practices.

05

Neighborhood Aggregation Pooling (Milk-Run Logistics)



Clusters small neighborhood lots into a bulk pickup batch and triggers an automated voice call to notify the collector.



FEASIBILITY AND VIABILITY

FEASIBILITY ANALYSIS

- **Operational Viability:** The pivot correctly targets the local scrap dealer as the primary user, leveraging their existing business volume and smartphone literacy to handle the actual QR handovers and EPR certificate generation.
- **Financial Viability:** Strong market adoption is driven by the October 2024 GST rules. Dealers now face an 18% reverse charge penalty for undocumented purchases; this platform's ledger directly shields them from these new financial hits.
- **Technical Viability:** A dual-layered approach is highly practical. It provides a robust compliance tool for the dealer while giving the individual door-to-door kabadiwala a lightweight, voice-first interface solely for fair price-checking and safety warnings.

POTENTIAL CHALLENGES & RISKS



Market Competition: Existing platforms (like Attero's MetalMandi and Recykal) already serve dealers with live commodity pricing.



Informal Market Inertia: Dealers and kabadiwalas may resist digitizing transactions to avoid government oversight or taxation.



Toxic Processing Habits: The informal sector currently resorts to burning cables and acid-leaching PCBs because it yields immediate cash, destroying valuable minerals in the process.



Volume Mismatch: Individual door-to-door kabadiwalas do not possess enough scrap to justify direct transactions with authorized recyclers.

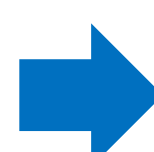
STRATEGIES FOR OVERCOMING CHALLENGES



Hyper-Specific Differentiation: The system is explicitly built for EPR compliance, capturing GPS, photos, and timestamps to generate government-ready certificates. It also reaches one layer lower with voice-first tools for illiterate users..



GST Penalty Avoidance: Position the platform not as regulatory surveillance, but as a financial shield that prevents dealers from absorbing the new 18% GST reverse charge.



Profitable Diversion: Integrate safety warnings into the kabadiwala app while providing dealers a streamlined, profitable channel to sell intact e-waste to formal recyclers instead of hazardous backyard processors.



Aggregator-Centric Hubs: Utilize the dealer as the primary aggregation node. Individual kabadiwalas use the app for price-checking, while dealers consolidate the volume needed to make recycler pickups profitable.

IMPACT AND BENEFITS

PLATFORM BENEFITS

1. **Price Visibility** - Reduces information asymmetry between informal sellers and buyers.

2. **Value Preservation** - Helps high-value e-waste reach the right grade and recycling pathway.

3. **Formal Market Access** - Connects informal collection with authorized recycling channels.

4. **Trust & Accountability** - verifiable record of material movement, handover and payment.

5. **Efficient Aggregation** - fragmented collections into organized, recycler-ready bulk lots.

6. **Inclusive Participation** - Brings the least-visible participant into the digital recycling ecosystem.

IMMEDIATE EFFECT

FOR KABADIWALA

Access to transparent price information helps kabadiwalas make better-informed selling decisions and avoid undervaluing high-value e-waste.

FOR DEALER/ AGGREGATOR

Digital lot creation, price benchmarks and recycler matching help dealers organize bulk material and make better procurement and selling decisions.

FOR THE AUTHORIZED RECYCLER

Traceable material records and verified handovers provide recyclers with documented information needed for formal recycling and EPR processes.

UNIT ECONOMICS & PLATFORM SUSTAINABILITY

₹38–₹1,960/kg: Listed PCB rates vary 50×+ by type/grade, highlighting the value of accurate classification and price discovery.

Proposed model: Technology/traceability fee from formal participants → collector-facing access remains free.

Source: MetalMandi, New Delhi rates observed Sep 2026; rates vary by type/grade.

LONG TERM IMPACT

ECONOMIC EMPOWERMENT

Improved price transparency and market access can strengthen bargaining power and earning potential

FORMALIZATION OF RECYCLING

Dig stronger bridge between informal collection and the authorized recycling ecosystem

ENVIRONMENTAL SUSTAINABILITY

Reduce reliance on unsafe processing and support safer material recovery.

SOCIAL INCLUSION AND SAFETY

Local-language, low-literacy interaction and safety guidance make participation more accessible while promoting safer handling

- **CPCB Official Rules Document:**
- **Registration SOP & Form Guidelines:**
- **On-Device Machine Learning Engine:**
Android Background Auto-Sync:
- **Offline Local Storage Database:**
- **Milk-Run Logistics Routing:**
- **Datsets:**

MoEFCC E-Waste Management Rules 2022 PDF

CPCB Standard Operating Procedure for Stakeholders

TensorFlow Lite MobileNet API Documentation

Android WorkManager Task Scheduling Guide

SQLite Documentation

arXiv: Vehicle Routing Problem Optimization Models

Balanced E-Waste Dataset Computer Vision Model
E Waste Image Dataset